SINKGRAB · REGULATORY AND STANDARDS REGISTER

SinkGrab regulatory and standards register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SKG-REG-001	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sinkgrab		
Source	docs/07-regulatory.md @ ec186d3		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	G1 points at PR-007 as restated by Amish's decision O4 of 2026-10-25 (proof at 1.25 x the rated load, slip clutch)
0.1	2026-10-24	Amish Chadha	First register (kit 1.9.1 upgrade); drinking-water and manual-handling entries reuse the Food and water security focus-area research of WellSense (WLS-REG-001) and BankHaul (BKH-REG-001)

SinkGrab regulatory and standards register

SinkGrab is hand-powered lifting gear worked over an open well, so the rules that matter most are those for lifting equipment, gas testing at shafts and manual handling, then the rules for working on a drinking-water well. Nothing here is certified: SinkGrab is an open engineering reference at TRL 3, and every entry marked "to assess" needs a qualified reviewer in the trial country before any claim is made. Not legal advice. Drinking-water and manual-handling entries are the focus-area research already done for WellSense and BankHaul, cited again here. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	EN 13157, cranes, safety, hand powered cranes	European Union; reference elsewhere	Design, static and dynamic tests and marking of hand-powered lifting equipment	The hand capstan, lead sheave, line and HatchSide tripod working together as one hand-powered hoist; its test coefficients against PR-007's 1.25 x the rated load (restated 2026-10-25, SKG-DEC-001 O4), and how it treats a slip clutch as the overload limiter	Design lead	To assess
G2	Machinery Directive 2006/42/EC, replaced by Machinery Regulation (EU) 2023/1230	European Union	Essential safety requirements for machinery, including lifting machinery moved by hand	Only if SinkGrab were supplied in the EU; not at TRL 3	Design lead	Does not apply now; to assess before any supply
G3	Lifting Operations and Lifting Equipment Regulations 1998 (LOLER)	United Kingdom (practice reference)	Thorough examination, proof testing and records for lifting equipment	Reference practice for the proof load and daily checks (BLD safety stops 2 and 9)	Design lead	Reference only
G4	EN 13889, forged steel shackles for general lifting, grade 6	European Union; reference elsewhere	Rated shackles and their marking	The swivel and shackles (BOM line 29)	Design lead	To assess (buy rated and marked)
G5	ISO 1141 (polyester fibre ropes), EN ISO 10547 (polyester double braid) and ISO 2307 (test methods)	International	Rope construction, minimum breaking force and its test	The 8 mm closing line, the recovery line and saddle lines (BOM lines 26 to 28); the 12 kN breaking strength assumed in SKG-CAL-001 E3	Design lead	Applies when buying: specify to EN ISO 10547 or equal and ask for the test certificate
G6	ANSI/ASSP A10.48 (via ASME B30.26 practice)	United States (practice reference)	Minimum sheave-to-rope ratio of 6:1 for double-braid rope	Sheaves and drum: 16.2 and 22.0 rope diameters (STEM S13)	Design lead	Met by design
G7	ISO 11228-1 and ISO 11228-2, manual handling (lifting and carrying; pushing and pulling)	International	Reference masses and force limits for manual work	Pieces of 25 kg or less (PR-009); the grab never lifted by hand (K3); the spoil tub slid, not lifted (K4); the T-bar and jerk-bar forces (STEM B4, OB5)	Design lead	Applies; B4 assessment to do
G8	EN 60079-29-1 (flammable gas	European Union;		The four-gas detector (BOM line 31) and the		To assess: buy detectors with a

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	detectors) and EN 45544 (toxic gas detectors)	reference elsewhere	Performance of portable gas detectors	carbon dioxide monitor (line 35) are bought, not made	Design lead	performance certificate and a calibration record
G9	OSHA 29 CFR 1910.146, permit-required confined spaces	United States (practice reference)	Testing, entry permits, attendants and rescue for confined spaces such as wells	SinkGrab's method forbids entry; any entry for another reason is a separate operation (BLD safety stop 8)	Co-design partner	Reference only
G10	WHO *Guidelines for Drinking-water Quality*, 4th edition incorporating the first and second addenda (2022)	International (guidance)	Microbial and chemical guidance, sanitary protection of groundwater sources	Context: SinkGrab adds nothing to the water; the well is cleaned and disinfected after the work	Co-design partner	Applies as guidance
G11	IS 10500:2012, drinking water specification, Bureau of Indian Standards	India	Limits for drinking water quality	Context for the Kerala trial: the water is tested after the work, as for any well	Co-design partner	Applies as context
G12	AWWA C654-13, disinfection of wells	United States (practice reference)	Disinfecting a well and the equipment put into it	Disinfecting the grab, scraper and well after the work (Oxfam guidance is the field reference)	Co-design partner	Reference only
G13	Kerala Ground Water (Control and Regulation) Act, 2002, and the Central Ground Water Authority guidelines of 24 September 2020	India (Kerala; pan-India where states do not regulate)	Permission to construct or deepen wells in notified areas	Deepening a well at the Kerala trial site	Co-design partner	To assess with the Kerala Ground Water Department
G14	IS 3955:1967, code of practice for design and construction of well foundations	India	Bridge well foundations: sinking, tilt (1 in 60) and shift limits	Reference only: SinkGrab's 1 in 80 tilt stop is stricter (PR-006)	Design lead	Reference only
G15	EN 10219 and EN 10025-2 (hollow sections; S235 and S275 structural steel), ISO 4017, ISO 4032, ISO 7089, ISO	International	Stock sections, steel grades and fastener dimensions	Every made part and fastener in the detailed model	Design lead	Met by design

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	2341 and ISO 1234 (fasteners and pins)					
G16	Local rules on lifting gear and work at open shafts at the trial site	Trial country	Inspection of lifting gear, barriers, gas tests	Both trial candidates (India, Ethiopia)	Co-design partner	To assess

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- Bureau of Indian Standards. *IS 10500:2012, Drinking Water, Specification (Second Revision)*. As cited in WellSense (WLS-REG-001).
- American Water Works Association. *AWWA C654-13, Disinfection of Wells*. As cited in WellSense (WLS-REG-001).
- Central Ground Water Authority. *Guidelines to Regulate and Control Ground Water Extraction in India*, notification S.O. 3289(E), 24 September 2020. As cited in WellSense (WLS-REG-001).
- World Health Organization. *Guidelines for Drinking-water Quality*. 4th ed. incorporating the first and second addenda. Geneva: WHO, 2022.
- Messer, Caleb. "Rope and Rigging Compatibility, Part Two of Two." *Inside Towers*, September 6, 2019. https://insidetowers.com/cell-tower-news-rope-and-rigging-compatibility-part-two-of-two/ (checked 2026-10-25).
- BSI. *BS EN ISO 10547:2009, Polyester Fibre Ropes, Double Braid Construction*. https://knowledge.bsigroup.com/products/polyester-fibre-ropes-double-braid-construction (checked 2026-10-25).
- Henderson, Bob. "Measuring Carbon Dioxide Levels." *Safety+Health*, September 1, 2010 (gas detection practice). https://www.safetyandhealthmagazine.com/articles/measuring-carbon-dioxide-levels-2 (checked 2026-10-25).
- Oxfam WASH. "Repairing, Cleaning and Disinfecting Hand Dug Wells." https://www.oxfamwash.org/repairing-cleaning-and-disinfecting-hand-dug-wells/.
- EN 13157, EN 13889, ISO 1141, ISO 2307, ISO 11228-1 and -2, EN 60079-29-1, EN 45544, OSHA 29 CFR 1910.146, LOLER 1998, Directive 2006/42/EC, Regulation (EU) 2023/1230, IS 3955:1967, EN 10219, EN 10025-2 and the ISO fastener standards are cited by number; editions and applicability are to assess where marked.